

Kiro Cheat Sheet

Spec-driven, agentic development. Kiro turns prompts into structured specs, then executes them with parallel agents. The spec is the source of truth. Code is the build artifact. Works across IDE, CLI, Web, and Mobile.

Install & Go

IDE: Download at kiro.dev. Open project, click Generate Steering Docs.

CLI: `curl -fsSL https://cli.kiro.dev/install | bash`

Steering Files — Project Context

Click Generate Steering Docs in the Kiro panel to auto-create `.kiro/steering/`. Add custom ones for coding standards, workflows, team practices.

Contains	Why It Matters
Product purpose + context	Kiro understands what it builds
Tech stack + frameworks	Correct patterns, no guessing
Project structure + conventions	Consistent with team standards
Custom rules (standards, workflows)	Team culture enforced by AI

Specs — Structured Feature Planning

Transform a prompt into a full implementation plan. Click Spec or + in the Specs section.

Phase	What Kiro Produces	Format
1. Requirements	User stories + acceptance criteria	EARS notation
2. Design	Technical architecture + components	Design doc (.md)
3. Tasks	Discrete, trackable implementation steps	Checklist (tasks.md)

Execute tasks by clicking them. Status auto-updates. Great for production features, refactors, understanding systems.

Two Ways to Work

	Vibe (Freeform)	Spec (Structured)
How	Prompt > code directly	Prompt > reqs > design > tasks > code
Best for	Prototypes, exploration, quick fixes	Production features, team collab
Traceability	Minimal	Full: requirement > design > code > test

Hooks — Event-Driven Automation (v1 JSON)

Stored in `.kiro/hooks/`. Create in natural language from panel (+).

Trigger	Example
File saved	"When I save a component, auto-update its test"
File created	"New route created > generate OpenAPI docs"
Pre/Post task execution	"Before task: check deps" / "After: run tests"
Manual / Agent stop	"On demand: lint + scan" / "Summarize what changed"

Custom Agents & Modes

Custom Agents: `.md` files in `.kiro/agents/`. Tag-based tools (read, write, shell, web), inline MCP + permissions. Switch mid-session. Ideas: security-reviewer, deployer, test-writer.

Supervised: Pauses for approval (starting out). **Autopilot:** Executes freely (flow state, trusted workflows).

Kiro Cheat Sheet (cont.)

Agent Focus Mode — Parallel Sessions

Agent-first layout (top-right toggle). Multiple parallel sessions, workflow picker (Spec / Plan / Bug Fix / freeform), auxiliary panel for specs + diffs alongside chat. Cloud Sessions in preview.

Kiro CLI — Agents in Your Terminal

Full Kiro agent power in terminal. Same steering files + MCP as IDE. Models: Claude Sonnet 4.5, Haiku 4.5, Auto. Custom agents work in CLI too. Credits shared across IDE and CLI.

Kiro Autonomous Agent — Cloud Frontier Agent

Persistent cloud agent that works independently across multiple repos. Maintains context across sessions and **learns from your code reviews**.

- Describe task > spins up sandbox > clones repos > coordinates sub-agents > opens PRs.
- Up to 10 concurrent tasks. Runs asynchronously while you work on other things.
- GitHub: add 'kiro' label to issues or mention /kiro in comments to assign work.
- Learns: PR feedback is remembered and applied to future tasks automatically.
- Configurable: network access levels, MCP integrations, env vars/secrets per task.

Property-Based Testing — Spec Correctness

Measures if code matches spec. Extracts properties from your requirements, generates hundreds of random test cases. Uses "shrinking" to find minimal counter-examples. Catches edge cases you'd never write manually.

Checkpointing & Permissions

Checkpointing: Auto-checkpoint on every agent action. Roll back any number of steps without losing progress.

Permissions: Capability-based. Default: reads + read-only git. Everything else prompts. Persist as 'always allow'/'always deny'. Scoped per-workspace or global.

MCP & Quick Start

MCP: AWS, PostgreSQL, GitHub, Brave Search. Configure from Kiro panel > MCP. Remote MCP supported.

Step	How
1. Open project	File > Open Folder or kiro .
2. Generate steering	Kiro panel > Generate Steering Docs
3. Create spec	Specs > + > describe feature
4. Set up hook	Agent Hooks > + > natural language
5. Add MCP	MCP > pencil > add servers
6. Custom agent	Save .md in .kiro/agents/
7. Agent Focus	Top-right toggle > parallel sessions

Power Patterns

- **Full feature:** Spec: "Add multi-tenant auth with SSO, RBAC, audit logging." Execute tasks.
- **Parallel:** "Subagents: 1) Map APIs 2) DB schema 3) Auth flow. Summaries only."
- **Auto-test hook:** "When I save in src/, run related test and report failures."
- **Deploy agent:** Deployer agent with shell + AWS MCP. "Deploy to staging via CDK."
- **Security agent:** Read-only reviewer. "Review changes for vulnerabilities."

Team: Specs = collaboration contract. .kiro/ is version-controlled and shared automatically. Export sessions (right-click > Export Chat). Organizations: IAM Identity Center, manage from AWS Console.